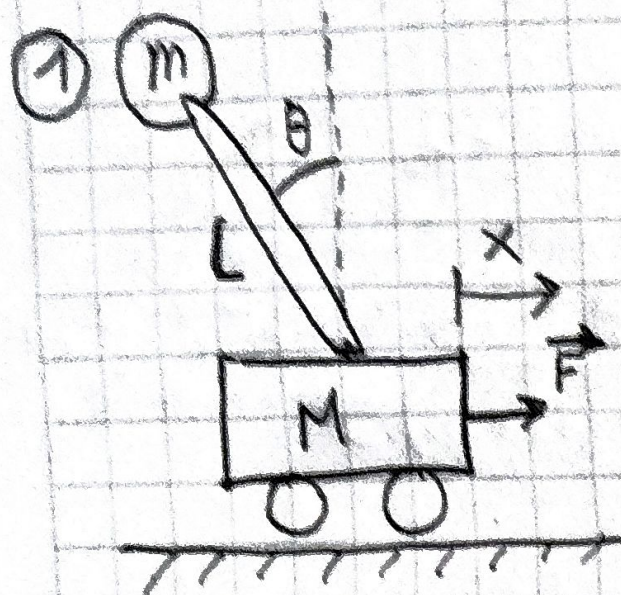




$$\textcircled{a} \quad \begin{aligned} X_m &= x + L \sin \theta \rightarrow \dot{X}_m = \dot{x} + L \cos \theta \dot{\theta} \\ Y_m &= L \cos \theta \rightarrow \dot{Y}_m = -L \sin \theta \dot{\theta} \end{aligned}$$

$$K = \frac{1}{2} M \dot{x}^2 + \frac{1}{2} m (\dot{X}_m^2 + \dot{Y}_m^2) =$$

$$= \frac{1}{2} (M+m) \dot{x}^2 + mL \cos \theta \dot{x} \dot{\theta} + \frac{1}{2} m L^2 \dot{\theta}^2$$

$$P = mgL(1 - \cos \theta).$$

$$L = K - P = \frac{1}{2} (M+m) \dot{x}^2 + mL \cos \theta \dot{x} \dot{\theta} + \frac{1}{2} m L^2 \dot{\theta}^2 - (mgL(1 - \cos \theta))$$

$$\frac{d}{dt} \left(\frac{\partial L}{\partial \dot{x}} \right) - \frac{\partial L}{\partial x} = (M+m) \ddot{x} + mL (\ddot{\theta} \cos \theta - \dot{\theta}^2 \sin \theta) = F \quad \textcircled{1}$$

$$\frac{d}{dt} \left(\frac{\partial L}{\partial \dot{\theta}} \right) - \frac{\partial L}{\partial \theta} = mL \ddot{x} \cos \theta + mL^2 \ddot{\theta} - mLg \sin \theta = 0 \quad \textcircled{2}$$

$$\begin{aligned} \theta &\approx 0 \\ \sin \theta &= \theta \\ \cos \theta &= 1 \end{aligned}$$

$$(M+m) \ddot{x} + mL \ddot{\theta} = F \quad \textcircled{3}$$

$$mL \ddot{x} + mL^2 \ddot{\theta} - mgL \theta = 0 \quad \textcircled{4}$$

$$\ddot{x} = \frac{1}{M} (F - mg\theta) \quad \textcircled{B}$$

$$\ddot{\theta} = \frac{1}{ML} ((M+m)g\theta - F)$$

$$\begin{bmatrix} \ddot{x} \\ \ddot{\theta} \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & -\frac{mg}{M} & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & \frac{(M+m)g}{ML} & 0 \end{bmatrix} \begin{bmatrix} x \\ \dot{x} \\ \theta \\ \dot{\theta} \end{bmatrix} + \begin{bmatrix} 0 \\ \frac{1}{M} \\ 0 \\ -\frac{1}{ML} \end{bmatrix} F$$

$$Y = \begin{bmatrix} 1 & 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} x \\ \dot{x} \\ \theta \\ \dot{\theta} \end{bmatrix} + \begin{bmatrix} 0 \end{bmatrix} F$$

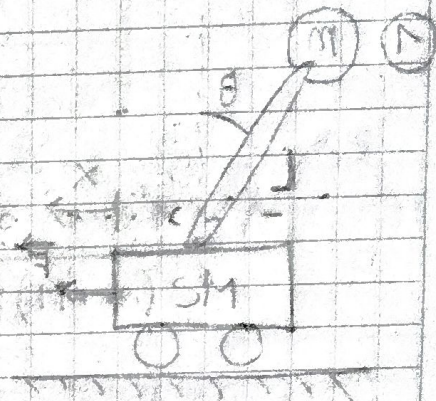
$$M = 1 \text{ Kg}; m = 0,1 \text{ Kg}; L = 0,5 \text{ m}; g = 9,806 \text{ m/s}^2$$

$$F(s) = \frac{s^2 - 19,61}{s^4 - 8,82 \times 10^{-16} s^3 - 21,57 s^2 - 21,582 s^2}$$

© $t_{sla} = \text{inf.}$ debido a la naturaleza del sistema, no se encuentra un tiempo de establecimiento definido; se tomará como tiempo de establecimiento de 1,25s.

$$\textcircled{d} \quad F(s) = \frac{s^2 - 19,62}{s^4 - 21,582 s^2} \quad \text{ess} = 0 \quad \zeta = 0,95 \quad W_n = \frac{4}{1,25(0,9)} = 3,52$$

$$S = -3,168 + 1,534j$$

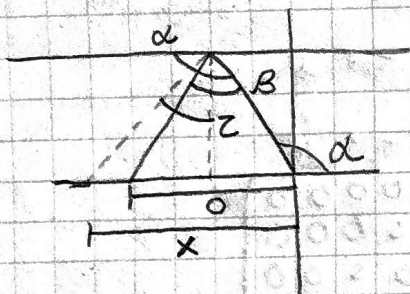


$$\frac{s^2 - 19,62}{s^4 - 21,582s^2} \Big|_{-3,168 + 1,534j} = \underbrace{4,109 \times 10^{-2}}_I + \underbrace{6,064 \times 10^{-2}j}_{II}$$

$$\tan^{-1}\left(\frac{6,064 \times 10^{-2}}{4,109 \times 10^{-2}}\right) = \underbrace{55,87}_I \rightarrow 180 - 55,87 = \underbrace{124,127}_II$$

$$124,12 / 3 = 41,37^\circ \approx 42^\circ$$

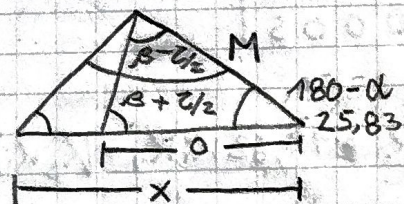
#2 $^\circ$ a complement



$$\alpha = \tan^{-1}\left(\frac{-1,534}{-3,168}\right) = -25,83$$

$$180 - 25,83 = 154,16 \alpha$$

$$\beta = \alpha/2 = 77,08 \quad \gamma/2 = 20,687^\circ$$



$$M = \sqrt{-3,168^2 + 1,534^2} = 3,52$$

$$\frac{\sin(56,39)}{0} = \frac{\sin(97,76)}{3,52}$$

$$0 = 2,958$$

$$\frac{\sin(97,76)}{x} = \frac{\sin(56,39)}{3,52}$$

$$x = 4,187$$

$$\frac{s^2 - 19,62}{s^4 - 21,582s^2} \left(\frac{K(s + 2,958)^3}{(s + 4,187)^3} \right) \Big|_{-3,168 + 1,534j} = (-4,35 \times 10^{-2} - 7,21 \times 10^{-13}j)K$$

$$K^2((-0,0435)^2 + (7,21 \times 10^{-13})^2) = 1$$

$$K = 22,987$$

$$A = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & -21,58 & 0 \end{bmatrix} \quad B = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 1 \end{bmatrix}$$

$$C = [-19,62 \quad 0 \quad 1 \quad 0] \quad D = [0]$$

$$BK = \begin{bmatrix} 0 & 0 & 0 & K_1 \\ 0 & 0 & 0 & K_2 \\ 0 & 0 & 0 & K_3 \\ 0 & 0 & 0 & K_4 \end{bmatrix}$$

$$A - BK = \begin{bmatrix} 0 & 1 & 0 & -K_1 \\ 0 & 0 & 1 & -K_2 \\ 0 & 0 & 0 & 1 - K_3 \\ 0 & 0 & -21,58 & -K_4 \end{bmatrix}$$

$$A^* = \begin{bmatrix} 0 & 1 & 0 & -K_1 & 0 \\ 0 & 0 & 1 & -K_2 & 0 \\ 0 & 0 & 0 & 1 - K_3 & 0 \\ 0 & 0 & -21,58 & -K_4 & K_i \\ 19,62 & 0 & -1 & 0 & 0 \end{bmatrix} \quad SI = \begin{bmatrix} s & 0 & 0 & 0 & 0 \\ 0 & s & 0 & 0 & 0 \\ 0 & 0 & s & 0 & 0 \\ 0 & 0 & 0 & s & 0 \\ 0 & 0 & 0 & 0 & s \end{bmatrix}$$

$$\det(SI - A) = S^5 + K_4 S^4 + (21,58 - 21,58 K_3) S^3 + (K_i + 19,62 K_i K_1 - K_3 K_i) S^2 + 19,62 K_2 K_i S - (19,62 K_i + 19,62 K_3 K_i)$$

$$P_{old} = S^5 + 102,4 S^4 + 3699 S^3 + 53640 S^2 + 248600 S + 414300$$

$$K_4 = 102,4$$

$$21,58 - 21,58 K_3 = 3699 \quad K_3 = -170,41$$

$$-19,62 K_i - 19,62 K_3 K_i = 414300 \quad K_i = 124,64$$

$$19,62 K_2 K_i = 248600 \quad K_2 = 101,65$$

$$K_i + 19,62 K_i K_1 - K_i K_3 = 53640 \quad K_1 = 13,198$$

$$\frac{S^2 - 19,61}{S^4 - 21,57 S^2} e^{-1s} = \frac{S^2 - 19,61}{S^4 - 21,57 S^2} (1 - 1s) = \frac{-S^3 + S^2 + 19,61 S - 19,61}{S^4 - 21,57 S^2}$$

$$A = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & -21,57 & 0 \end{bmatrix} \quad C = [-19,61 \quad 19,61 \quad 1 \quad -1]$$

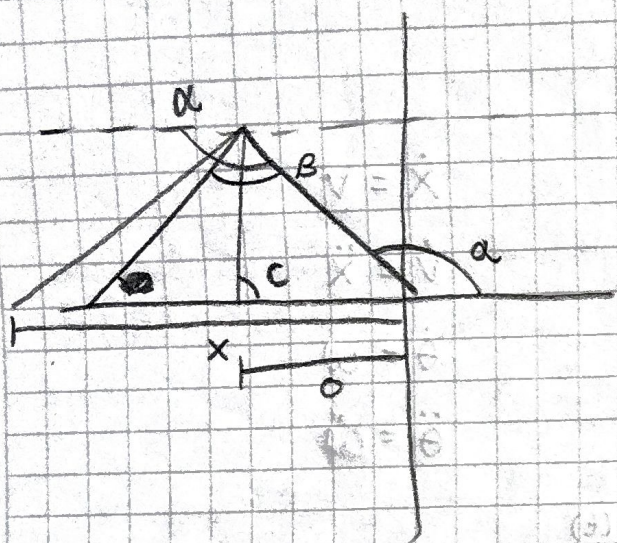
$$\frac{-s^3 + s^2 + 19,62s - 19,62}{s^4 - 21,57s^2}$$

$$= \frac{0,264 + 0,189j}{(-3,168 + 1,533j)}$$

$$\tan^{-1}\left(\frac{0,189}{0,264}\right) = 35,59^\circ$$

$$180 - 35,59 = 144,36^\circ$$

#Z



$$Z/2 = 24,05^\circ$$

$$\alpha = \tan^{-1}\left(\frac{1,533}{-3,168}\right) = 154,15^\circ$$

$$B = \alpha/2 = 77,075^\circ$$

$$180 - \alpha = 25,84^\circ$$

$$B - Z/2 = 53,01^\circ$$

$$C = 180 - 25,84 - 53,01 = 101,13^\circ$$

$$M = \sqrt{(3,168^2 + 1,533^2)} = 3,52$$

$$\frac{\sin(53,01)}{0} = \frac{\sin(101,13)}{3,52}$$

$$o = 2,866$$

$$x = 4,323$$

$$\frac{-s^3 + s^2 + 19,62s - 19,62}{s^4 - 21,57s^2} \left(\frac{K((s + 2,866)^2)}{(s + 4,323)^3} \right) = \frac{-0,175 - 4,073 \times 10^{-16}}{-3,168 + 1,533j}$$

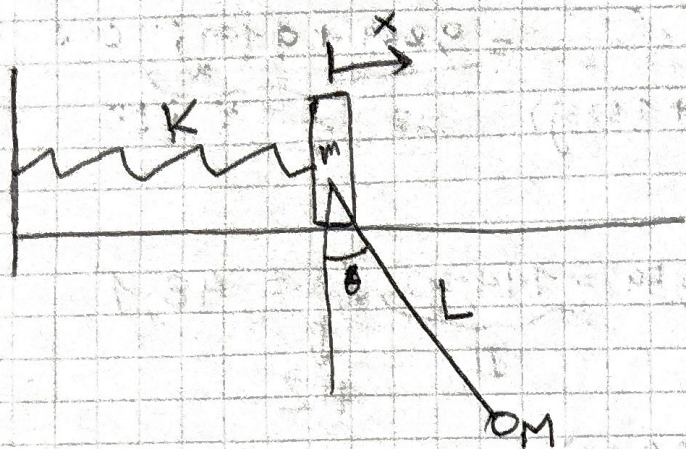
$$K^2(0,175^2 + 4,073 \times 10^{-16}) = 1$$

$$K = 5,6918$$

$$A^* = \begin{bmatrix} 0 & 1 & 0 & -K_1 & 0 \\ 0 & 0 & 1 & -K_2 & 0 \\ 0 & 0 & 0 & 1-K_3 & 0 \\ 0 & 0 & -21,58 & -K_4 & K_i \\ 19,62 & -19,62 & -1 & 1 & 0 \end{bmatrix}$$

$$\det(SI - A^*) = s^5 + K_4 s^4 + (21,58 - K_i - 21,58 K_3) s^3 + (K_i + 19,62 K_1 K_i - 19,62 K_2 K_i - K_3 K_i) s^2 + (19,62 K_i + 19,62 K_2 K_i - 19,62 K_3 K_i) s + (-19,62 - 19,62 K_3 K_i)$$

$$Pold = s^5 + 102,4 s^4 + 3699 s^3 + 53640 s^2 + 248600 s + 414300$$



$$\theta \approx 0$$

$$\cos \theta = 1$$

$$\sin \theta \approx \theta$$

$$(m+M)\ddot{x} + ML\ddot{\theta} + Kx = F(t)$$

$$ML\ddot{x} + ML^2\ddot{\theta} + MgL\theta = 0$$

$$\ddot{x} = -\frac{K}{m}x + \frac{Mg}{m}\theta + \frac{1}{m}F(t)$$

$$\dot{x} = v$$

$$\dot{v} = \ddot{x}$$

$$\dot{\theta} = \omega$$

$$\dot{\omega} = \ddot{\theta}$$

$$\ddot{\theta} = \frac{K}{Lm}x - \frac{(m+M)g}{Lm}\theta - \frac{1}{Lm}F(t)$$

$$\begin{bmatrix} \dot{x} \\ \dot{v} \\ \dot{\theta} \\ \dot{\omega} \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 & 0 \\ -\frac{K}{m} & 0 & \frac{Mg}{m} & 0 \\ 0 & 0 & 0 & 1 \\ \frac{K}{Lm} & 0 & -\frac{(m+M)g}{Lm} & 0 \end{bmatrix} \begin{bmatrix} x \\ v \\ \theta \\ \omega \end{bmatrix} + \begin{bmatrix} 0 \\ \frac{1}{m} \\ 0 \\ -\frac{1}{Lm} \end{bmatrix} F(t)$$

$$Y = \begin{bmatrix} 0 & 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} x \\ v \\ \theta \\ \omega \end{bmatrix} + \begin{bmatrix} 0 \end{bmatrix} F(t)$$

$$F(s) = \frac{\Theta(s)}{F(s)} = \frac{s^2}{Lm s^4 + (LK + (m+M)g)s^2 + gK}$$

$$m = 0,5 \text{ Kg}; M = 0,2 \text{ Kg}, L = 0,6 \text{ m}, K = 80 \text{ N/m}, g = 9,806 \text{ m/s}^2$$

$$F(s) = \frac{s^2}{0,30 s^4 + 54,867 s^2 + 704,8} = \frac{3,333 s^2}{s^4 + 182,89 s^2 + 2616}$$

$$\frac{3,333 S^2}{S^4 + 182,89 S^2 + 2616}$$

$$ess = 0$$

$$\downarrow$$

$$\left(\frac{1}{s}\right)^2 = \frac{1}{s^2}$$

$$tsd = 1s \quad \zeta = 0,7$$

$$\omega_n = \frac{4}{0,7} = 5,714$$

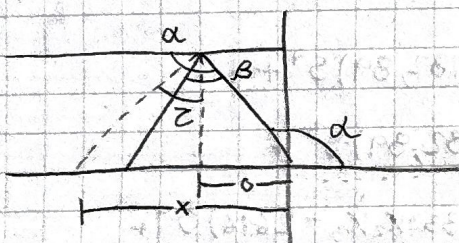
$$s = -\zeta \omega_n \pm j \omega_n \sqrt{1 - \zeta^2}$$

$$= -4 \pm j 4,080$$

$$\left. \frac{3,333 S^2}{S^2(S^4 + 182,89 S^2 + 2616)} \right|_{-4 + 4,080j} = 0,000128 + 0,000531j$$

I

$$\tan^{-1} \left(\frac{0,000531}{0,000128} \right) = \underbrace{76,44^\circ}_I \rightarrow 180 - 76,44 = 103,56^\circ / 2 = \underline{51,8^\circ}$$



$$\alpha = \tan^{-1} \left(\frac{4,080}{-4} \right) = \underbrace{-45,56^\circ}_{IV} + 180 = \underline{134,43^\circ}_{II}$$

$$\beta = \alpha / 2 = 67,22^\circ$$

$$U/2 = 25,9^\circ$$

$$M = \sqrt{4^2 + 4,08^2} = \underline{5,71}$$

$$\frac{\text{Sen}(41,31)}{0} = \frac{\text{Sen}(93,10)}{5,71}$$

$$\underline{0 = 3,778}$$

$$\frac{\text{Sen}(93,10)}{x} = \frac{\text{Sen}(41,31)}{5,71}$$

$$\underline{x = 8,641}$$

$$\left. \frac{3,333 S^2}{S^2(S^4 + 182,89 S^2 + 2616)} \left(\frac{K(S + 3,778)^2}{(S + 8,641)^2} \right) \right|_{-4 + 4,08j} = (-0,000239 + 8,96 \times 10^{-20}j)K$$

$$K^2(-0,000239^2 + (8,96 \times 10^{-20})^2) = 1$$

$$\underline{K = 4184,471}$$

$$3,333 S^2$$

$$ess = 0$$

$$z_d = 0,7 \quad \omega_{nd} = 5,714$$

$$S^4 + 182,89 S^2 + 2616$$

$$t_{sd} = 1s$$

$$A = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 2616 & 0 & 182,89 & 0 \end{bmatrix}; B = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 1 \end{bmatrix}$$

$$BK = \begin{bmatrix} 0 & 0 & 0 & K_1 \\ 0 & 0 & 0 & K_2 \\ 0 & 0 & 0 & K_3 \\ 0 & 0 & 0 & K_4 \end{bmatrix}$$

$$C = [0 \quad 0 \quad 3,333 \quad 0]; D = [0]$$

$$A - BK = \begin{bmatrix} 0 & 1 & 0 & -K_1 \\ 0 & 0 & 1 & -K_2 \\ 0 & 0 & 0 & 1 - K_3 \\ 2616 & 0 & 182,89 & -K_4 \end{bmatrix}$$

$$A^* = \begin{bmatrix} 0 & 1 & 0 & -K_1 & 0 & 0 \\ 0 & 0 & 1 & -K_2 & 0 & 0 \\ 0 & 0 & 0 & 1 - K_3 & 0 & 0 \\ 2616 & 0 & 182,89 & -K_4 & K_2 & K_1 \\ 0 & 0 & 0 & 0 & 1 & 1 \\ 0 & 0 & -3,333 & 0 & 0 & 0 \end{bmatrix} = SI(6)$$

$$\det(SI - A^*) = S^6 + (K_4 - 1)S^5 + (2616K_1 + 182,89K_3 - K_4 - 182,89)S^4 + (2616K_2 - 2616K_1 - 182,89K_3 - 3,333K_1K_3 + 182,89)S^3 + (2616K_3 - 2616K_2 - 3,333K_1 + 3,333K_1K_3 - 3,333K_2K_3 - 2616)S^2 + (2616 - 2616K_3)S + 0$$

$$2616 - 2616K_3 = 8368000 \quad K_3 = -3197,77$$

$$K_4 - 1 = 134,4 \quad K_4 = 135,4$$

$$2616K_1 + 182,89K_3 - K_4 - 182,89 = 6976 \quad K_1 = 226,35$$

$$2616K_2 - 2616K_1 - 182,89K_3 - 3,333K_1K_3 + 182,89 = -172000 \quad K_2 = -853,73$$

Taylor

$$\frac{3,333 S^2}{S^4 + 182,82 S^2 + 2616} e^{-3s} = \frac{3,333 S^2 (1 - 3s)}{S^4 + 182,82 S^2 + 2616} = \frac{-9,999 S^3 + 3,333 S^2}{S^4 + 182,82 S^2 + 2616}$$

$$C = [0 \quad 0 \quad 3,333 \quad -9,999]$$

$$A^* = \begin{bmatrix} 0 & 1 & 0 & -K_1 & 0 & 0 \\ 0 & 0 & 1 & -K_2 & 0 & 0 \\ 0 & 0 & 0 & 1 - K_3 & 0 & 0 \\ 0 & 0 & 182,89 & -K_4 & K_2 & K_1 \\ 0 & 0 & 0 & 0 & 1 & 1 \\ 0 & 0 & -3,333 & 9,999 & 0 & 0 \end{bmatrix}$$

El determinante de $SI - A^*$ no presenta cambios, por lo que las constantes se mantienen iguales.

$$3,333 S^2$$

$$S^4 + 182,89 S^2 + 2616$$

$$A = \begin{bmatrix} 0 & 0 & 0 & 2616 \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 182,89 \\ 0 & 0 & 1 & 0 \end{bmatrix}$$

$$B = \begin{bmatrix} 0 \\ 0 \\ 3,333 \\ 0 \end{bmatrix}$$

$$L = \begin{bmatrix} L_1 \\ L_2 \\ L_3 \\ L_4 \end{bmatrix}$$

$$C = [0, 0, 0, 1] \quad D = [0]$$

$$LC = \begin{bmatrix} 0 & 0 & 0 & L_1 \\ 0 & 0 & 0 & L_2 \\ 0 & 0 & 0 & L_3 \\ 0 & 0 & 0 & L_4 \end{bmatrix}$$

$$A - LC = \begin{bmatrix} 0 & 0 & 0 & 2616 - L_1 \\ 1 & 0 & 0 & -L_2 \\ 0 & 1 & 0 & 182,89 - L_3 \\ 0 & 0 & 1 & -L_4 \end{bmatrix}$$

$$SI = \begin{bmatrix} S & 0 & 0 & 0 \\ 0 & S & 0 & 0 \\ 0 & 0 & S & 0 \\ 0 & 0 & 0 & S \end{bmatrix}$$

$$\det(SI - (A - LC)) = S^4 + L_4 S^3 + (L_3 - 182,89) S^2 + L_2 S + (L_1 - 2616)$$

$$Pol d = S^4 + 70,4 S^3 + 1446 S^2 + 7363 S + 12950$$

$$\underline{70,4 = L_4}$$

$$L_3 - 182,89 = 1446 \quad \underline{L_3 = 1628}$$

$$\underline{L_2 = 7363}$$

$$L_1 - 2616 = 12950 \quad \underline{L_1 = 15566}$$